KARTIK MISHRA

Kartik876.km@gmail.com | +91-8826118975 | LinkedIn | Github | Portfolio

EDUCATION

Amity University, Noida, India **Aug 2017 – Mar 2021**
B.Tech. Biotechnology CGPA: 8.15/10

WORK EXPERIENCE

BHARTI AIRTEL LIMITED **Gurugram, India**

One of India's largest telecom providers serving 600M+ customers with 21.56B USD revenue and 24,420 employees.

Machine Learning Engineer **Aug 2025 - Present**

- Engineered the backend orchestration for an AI voice bot assisting broadband field engineers, developing asynchronous FastAPI microservices and a MongoDB data layer for ticket ingestion, priority call queueing, and centralized state management.
- Building an end-to-end conversational AI stack using LangGraph decision flows and knowledge bases, integrating low-latency STT/TTS and LLM microservices with Airtel IQ telephony via WebSockets.
- Implemented a Docker based Apache Airflow infrastructure, introducing the team's first standardized, CI/CD-enabled orchestration framework to automate pipeline scheduling, monitoring, and failure recovery.
- Revamped the WCL technology stack (2G, 4G, 5G) - migrating pipelines between PySpark and pandas based on workload profile, orchestrating 20+ ETL workflows via Airflow as team SME, reducing runtime by 75% and compute costs by 85%.
- Architected and deployed large-scale PySpark based data pipelines processing terabytes of broadband customer data daily across India, generating network health KPIs, churn prediction signals, and customer experience metrics for Airtel's flagship Broadband Experience initiative.
- Spearheading the optimization and maintenance of the Duct operations across 12 states in India - acting as tech lead and primary SME/POC for cross-functional stakeholders nationwide.
- Designed and deployed a real-time monitoring dashboard for Duct operations - tracking multi-server processes, completely eliminating manual inspections and providing stakeholders with instant operational visibility

MOODY'S INSURANCE SOLUTIONS **Noida, India**

Provides catastrophe models and risk intelligence for insurers, with 7.09B USD revenue and a workforce of 15,838.

Associate Analytics, Model Analyst **Apr 2023 - Aug 2025**

- Architected an end-to-end data orchestration framework on Dagster, integrating high-frequency weather data from 30+ global sources with CI/CD automation via Jenkins and GitHub Actions, improving pipeline efficiency by 90%.
- Implemented a Django-based data visualization dashboard using Mapbox, JavaScript, MongoDB, and Python, enabling global wind station tracking and cross-referencing with the Postgres database to identify over 18,000 missing stations
- Collaborated with the Event Response team to automate Support Center uploads via an API, reducing manual upload time from 40-60 minutes to zero, enhancing efficiency for natural disaster reports and forecasts
- Unlocked Visibility into HWind processes by building insightful statistics dashboard using JavaScript, Python, and Django, tracking daily and weekly data feed counts with interactive charts, identifying any problematic data transactions and feed downtimes.

VERISK ANALYTICS (Wood Mackenzie) **Gurugram, India**

Global insight business for renewable energy, & natural resources, with 1.9B USD revenue and 2000+ employees.

Data Associate **Nov 2021 - Apr 2023**

- Enhanced data collection and processing pipelines in Python on AWS SageMaker, improving efficiency by 60%
- Built Python automation scripts to extract oil price data from country reports, update Excel deliverables, and trigger email notifications on completion, reducing end-to-end processing time by 90%.
- Led project handover sessions with different teams, managed project roadmaps, story allocation, and task division using JIRA
- Crafted a Power BI dashboard to monitor JIRA tickets and track completion statistics, fostering team accountability and ensuring timely task fulfillment; implemented JIRA tickets for day-to-day tasks assigned to individual team members

PUBLICATIONS

- Identification of vaccine targets & design of vaccine against SARS-CoV-2 coronavirus using computational and deep learning-based approaches - PeerJ, May 2022
- Identification of Vaccine Targets in Pathogens and Design of a Vaccine Using Computational Approaches - Nature Scientific Reports, September 2021
- To Study the Effect of Unconventional Treatment Protocol on COVID-19 Patients in Delhi Using Artificial Intelligence Based Methods - SSRN, September 2020

EXTRACURRICULAR ACTIVITIES

Member & AVP Education, Toastmasters Club, West Delhi - Organized weekly meetings, speaker selection, and onboarded new members (Apr 2023 – Mar 2025)

Volunteer & Drive Lead, Robin Hood Army, Delhi - Led 30+ food distribution drives and taught 25+ students weekly (Oct 2022 – Present)

Coordinator, CRC Placement Cell, Amity University - Orchestrated 100+ on-campus placement drives and pre-placement prep sessions (Feb 2019 – Mar 2021)

New Delhi, India - 110059

SKILLS

Languages: Python, SQL, JavaScript
Big Data: Apache PySpark, Pandas, Hadoop, HDFS, YARN, Delta Lake
Orchestration: Apache Airflow, Docker, Docker Compose, Dagster, Jenkins, GitHub Actions
ML/AI: PyTorch, Transformers, HuggingFace, LangGraph, LangChain, LLMs, STT/TTS (Whisper)
Backend & Web: FastAPI, Streamlit, WebSockets, Django
Databases: PostgreSQL, MongoDB
Visualization: Django, Power BI, Tableau
Tools: Power BI, Tableau, Git, Linux, Paramiko, AWS SageMaker
Spoken: English (Proficient), Hindi (Native)